



Incidence of Red Rot and its Impact on Quality Attributes in Bihar, India

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ABSTRACT

Red rot is one of the most devastating disease of sugarcane caused by *Colletotrichum falcatum*. The survey was carried out in different sugarcane growing areas of Bihar during 2019-20 and the observations were recorded. On the basis of observations, the highest disease incidence observed in Riga (23.32%) followed by Manjhaulia (19.90%), Lauriya (18.82%), Sidhwalia (15.83%), Sugauli (15.61%), Gopalganj (12.45%), Narkatiaganj (11.58%), Harinagar (6.30%), Pusa (6.04%) and lowest disease incidence (5.22%) in Hasanpur areas. However, the variety wise survey showed highest disease incidence in the variety CoH 160 (27.60%) followed by Co 0238 (24.02%), BO 128 (22.38%), CoH 167 (16.97%), Co 0233 (15.47%), CoSe 92423 (12.13%), CoSe 95422 (9.45%), CoJ 64 (8.50%), CoS 8436 (8.34%), CoLk 8102 (7.81%) and BO 145 (6.64%) while, lowest in BO 138 (4.50%). The three red rot isolates (CF 01, CF 07 and CF 08) tested on various sugarcane varieties exhibited BO 153 (2.50), BO 91 (2.70), BO 154 (2.93) and CoP 2061 (3.30) varieties were moderately resistant while, CoS 8436 (6.67) and CoSe 95422 (7.03) showed susceptible response to the disease. Furthermore, the highest reduction in juice qualities was noticed in case of CF 08 isolate followed by CF 07 and CF01 respectively.

Keywords – Red rot, survey, incidence, isolates, juice qualities, Bihar

INTRODUCTION

Sugarcane (*Saccharum officinarum* L.) is an important and unique cash crop. It is grown in both tropical and sub-tropical regions of the country and chief source of sugar providing 60% of the total sugar requirement of the country. In the country, it occupies an area of 5.2 million hectares with production of 400.37

million tonnes and productivity of 76.99 tonnes/ha. In Bihar state it is cultivated in an area of 3.04 lakhs ha, production of 182.85 lakh tonnes and productivity of 60.15 tonnes/ha (Government of Bihar, 2019). It is long duration crop and remains in the field for 12 to 16 months and subjected to attack of many plant pathogens. Among the various factors responsible for crop failure and decline in sugar production epiphytotic diseases have been considered to play an important role. Due to introduction of high sugared and high yielding varieties of cane in various parts of the country in recent years, association of diseases with different insect pests have become limiting factor in production and reduction in the yield considerably. Bihar is considered to be hot spot for sugarcane diseases due to favourable climatic condition particularly temperature and humidity. More than 20 diseases of sugarcane caused by different groups of pathogens have been reported from Bihar. The diseases like red rot, wilt, smut, pokkah boeng, sett rot, mosaic, grassy shoot, red stripe, and ratoon stunting diseases are limiting factors in sugarcane production in Bihar, among them the red rot caused by *Colletotrichum falcatum* is the most serious disease. Barber (1901) observed and reported first time the occurrence of red rot in India. Since then various red rot epidemics have been reported from different parts of India and resulted in failure of several popular varieties from cultivation (Singh, 2008; Babu *et al.*, 2009 and Babu, 2010). The estimated loss in crop production due to fungal diseases in India is about 18-31 per cent (Jayashree *et al.*, 2010) and it causes severe losses in yield and quality of the susceptible varieties in the Indian subcontinent (Duttamajumder, 2008). Due to red rot reduction in cane weight upto 29% and loss in sugar recovery by 31% was observed by Hussnain and Afghan (2006). The popular varieties are becoming

susceptible to red rot infection due to emergence of new races of the *Colletotrichum falcatum* and leading to varietal breakdown in Bihar. The present investigation was carried out to know the varietal susceptibilities and status of red rot disease naturally occurring in cane growing areas of Bihar during 2019-20.

MATERIALS AND METHODS

Survey

An extensive survey was carried out in various reserved areas of sugar factories of Bihar existing in different districts viz. Riga (Sitamarhi), Manjhaulia, Lauriya, Harinagar, Narkatiaganj (West Champaran), Sidhwalia, Gopalganj (Gopalganj), Pusa, Hasanpur (Samastipur) and Sugauli (East Champaran). Ten plots in each area were surveyed and observations were recorded from hundred canes which were randomly selected from four different sites in each plot aged between 7 to 10 months old crop. The red rot incidence percentage was calculated by the given formula:

$$\text{Disease incidence (\%)} = \frac{\text{Number of affected canes}}{\text{Total number of canes examined}} \times 100$$

The canes were examined for the red rot disease symptoms i.e. golden yellow leaves, drying of top portion, on splitting the internal tissues emit peculiar sour odour and reddish brown lesions with white spots were noticed.

Disease Incidence and Qualitative Parameters

The three budded setts of six sugarcane varieties namely BO 153, BO 91, BO 154, CoP 2061, CoS 8436 and CoSe 95422 were planted at Sugarcane Research Farm, Pusa (Bihar) during second week of February. Twenty canes of each variety was artificially inoculated by plug method with three red rot isolates i.e. CF 01, CF 07 and CF 08 in third week of August. The disease incidence was recorded according to above given formula. The juice quality parameters were estimated in lab by taking juice from the diseased canes. The brix was measured with brix hydrometer, sucrose with polariscope and purity coefficient was calculated using the given formula -

$$\text{Purity coefficient} = \frac{\text{Sucrose \%}}{\text{Corrected brix \%}} \times 100$$

RESULTS AND DISCUSSION

The sugarcane varieties exhibited various symptoms of the red rot disease during their growth stages. The drying of the top portion of the canes was observed prior to monsoon period that became more severe after the onset of monsoon. The top of the plants completely dried which indicated some sort of red rot infection. The symptoms become more prominent at later stages displaying discolouration and reddish lesion on the midrib of leaves along with drying of third and fourth spindle leaves. Infected leaves later turned to straw coloured and showed presence of black acervuli. The longitudinal splitting of the canes exhibited reddening of internodal tissues intermingled with the white bands. The white mycelial growth of the fungus was observed in some infected canes. Similar symptoms have been observed by several workers like Agnihotri and Singh (1977), Viswanathan *et al.* (2011), Jayashree *et al.* (2010) and Duttamajumder (2008). The alcoholic sour smell was emitted from the infected canes, which may be due to conversion of sucrose into glucose and alcohol (Duttamajumder, 2008).

The data depicted in the Table 1 clearly revealed that the highest disease incidence was observed in Riga (23.32%) followed by Manjhaulia (19.90%), Lauriya (18.82%), Sidhwalia (15.83%), Sugauli (15.61%), Gopalganj (12.45%), Narkatiaganj (11.58%), Harinagar (6.30%), Pusa (6.04%) and lowest disease incidence (5.22%) was recorded in Hasanpur areas during the year 2019-20. The data presented in Table 2 elucidated that regarding variety wise, the highest diseases incidence was reported in variety CoH 160 (27.60%) which was followed by Co 0238 (24.02%), BO 128 (22.38%), CoH 167 (16.97%), Co 0233 (15.47%), CoSe 92423 (12.13%), CoSe 95422 (9.45%), CoJ 64 (8.50%), CoS 8436 (8.34%), CoLk 8102 (7.81%) and BO 145 (6.64%) while, lowest was seen in variety BO 138 (4.50%).

Similar study had conducted by Kumar *et al.* (2018) they surveyed the incidence of red rot disease in variety CoS 8436 in Bihar. They reported loss in 147084 quintals in cane yield due to red rot infection during 2015-2018. Singh *et al.* (2021) also conducted a survey of sugarcane growing areas of Uttar Pradesh during 2015 to 2021. They observed severe incidence of red rot disease varying from 70 to 100% in central U.P.

Table 1: Incidence percentage of sugarcane red rot disease in different locations of Bihar

Location	Variety	Red Rot Incidence (%)
Riga	CoSe 92423	20.00
	CoLk 8102	7.81
	CoH 160	32.84
	Co 0233	25.60
	Co 0238	30.28
	Mean	23.31
Manjhaulia	CoJ 64	12.72
	CoS 8436	10.29
	CoH 167	18.69
	Co 0238	35.46
	CoH 160	22.36
	Mean	19.90
Lauriya	BO 128	22.38
	CoH 167	15.25
	Mean	18.82
Pusa	BO 145	6.64
	CoSe 92423	8.00
	BO 138	4.50
	CoSe 95422	5.00
	Mean	6.04
Sidhwalia	CoSe 95422	16.26
	CoSe 92423	15.40
	Mean	15.83
Sugauli	Co 0238	15.61
	Mean	15.61
Gopalganj	Co 0238	18.52
	CoS 8436	6.38
	Mean	12.45
Hasanpur	Co 0233	5.34
	CoSe 92423	5.10
	Mean	5.22
Harinagar	CoSe 95422	6.30
	Mean	6.30
Narkatiaganj	CoSe 95422	10.25
	Co 0238	20.22
	CoJ 64	4.27
	Mean	11.58

Table 2: Incidence percentage of red rot disease in different varieties grown in Bihar

Variety	Location	Red Rot Incidence (%)
CoSe 95422	Pusa	5.00
	Sidhwalia	16.26
	Harinagar	6.30
	Narkatiaganj	10.25
	Mean	9.45
CoSe 92423	Riga	20.00
	Pusa	8.00
	Sidhwalia	15.40
	Hasanpur	5.10
	Mean	12.13
Co 0233	Riga	25.60
	Hasanpur	5.34
	Mean	15.47
Co 0238	Riga	30.28
	Manjhaulia	35.46
	Sugauli	15.61
	Gopalganj	18.52
	Narkatiaganj	20.22
	Mean	24.02
CoS 8436	Manjhaulia	10.29
	Gopalganj	6.38
	Mean	8.34
CoJ 64	Manjhaulia	12.72
	Narkatiaganj	4.27
	Mean	8.50
CoH 160	Riga	32.84
	Manjhaulia	22.36
	Mean	27.60
CoH 167	Manjhaulia	18.69
	Lauriya	15.25
	Mean	16.97
CoLk 8102	Riga	7.81
	Mean	7.81
BO 128	Lauriya	22.38
	Mean	22.38
BO 138	Pusa	4.50
	Mean	4.50
BO 145	Pusa	6.64
	Mean	6.64

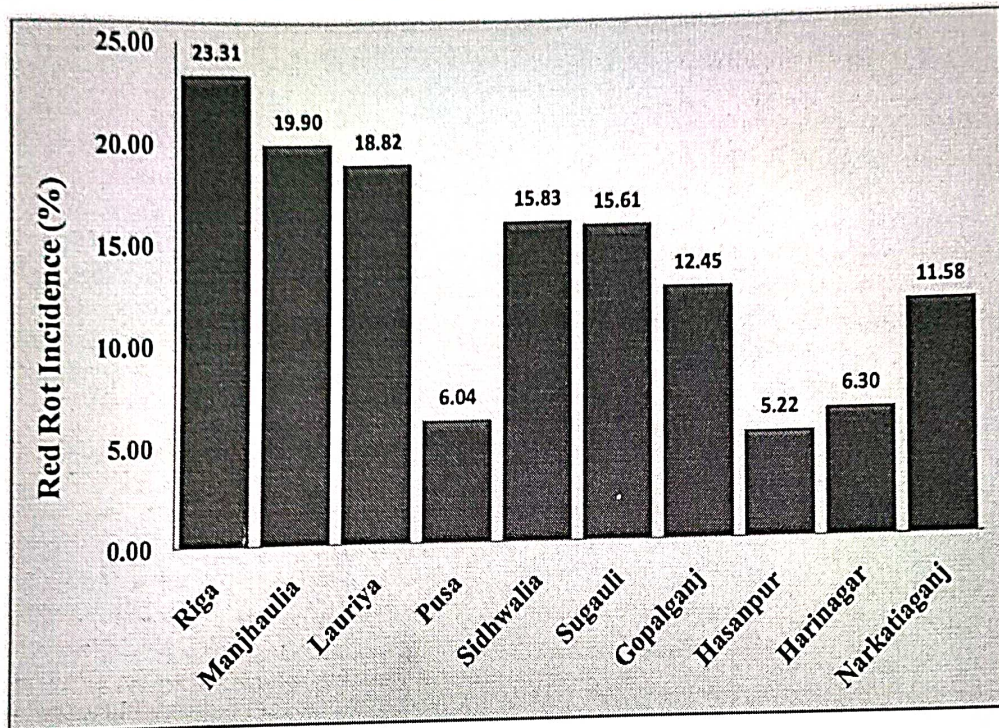


Fig 1: Incidence percentage of sugarcane red rot disease in different locations of Bihar

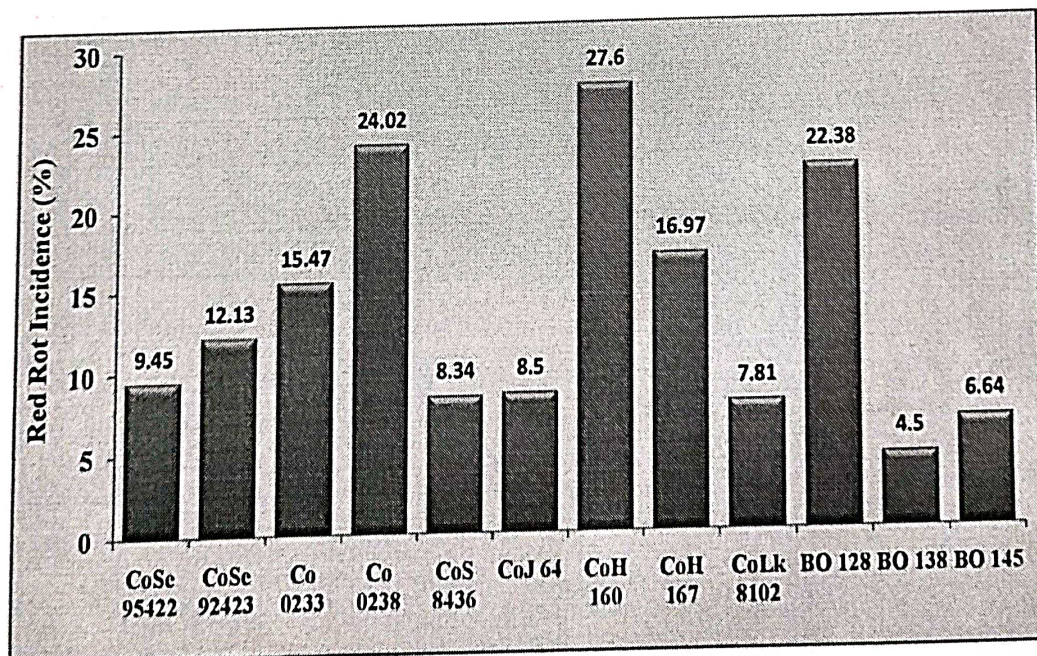


Fig 2: Incidence percentage of red rot disease in different varieties grown in Bihar

To know the effect of different isolates, six sugarcane varieties namely BO 153, BO 91, BO 154, CoP 2061, CoS 8436 and CoSe 95422 were inoculated with three (CF 01, CF 07 and CF 08) isolates of pathogen and disease was rated on 0-9 scale. The isolate CF 08 was highly virulent with disease score of 4.68 followed CF 07 (4.20) and CF 01 (3.68). In respect of the varieties, BO 153 (2.50) followed by BO 91 (2.70), BO 154 (2.93) and CoP 2061 (3.30) respectively showed their disease reaction as moderately resistant towards the red rot disease. While, the varieties CoS 8436 (6.67) and CoSe 95422 (7.03) were found susceptible towards the red rot disease infection.

The highest reduction in juice purity was noticed in case of CF 08 isolate followed by CF 07 and CF 01. The variety BO 153 displayed highest purity percentages (87.77%) followed by 85.93% and 81.23% for the isolate CF 01, CF 07 and CF 08 respectively. The variety BO 91 exhibited highest purity percentage of 86.70% for isolate CF 01, followed by 83.58% for isolate CF 07 while, lowest purity percentage of 78.07% for the isolate CF 08. Similarly, variety BO 154 showed purity percentages of 84.93%, 81.52% and 76.21% for isolates CF 01, CF 07 and CF 08 respectively. The purity percentages of 83.20%, 79.97% and 74.84% was

exhibited by variety CoP 2061 for the isolates CF 01, CF 07 and CF 08 respectively. The susceptible variety CoS 8436 displayed maximum purity percentage of 75.17%, 72.28% and minimum purity percentage 66.83% for the isolates CF 01, CF 07 and CF 08 respectively. The susceptible variety CoSe 95422 showed highest purity of 69.20% in case of isolate CF 01 followed by 66.36% for isolate CF 07 and 60.44% for the isolate CF 08. The highest purity percentage (87.77%) was recorded for resistant variety BO 153 and least virulent isolate CF 01 while, lowest purity percentage (60.44%) was exhibited by highly virulent isolate CF 08 in most susceptible variety CoSe 95422.

The findings regarding reduction in cane juice quality are also in close conformity with the earlier work of Minnatullah *et al.* (2012) they reported considerable reduction in brix, sucrose and purity of cane juice. The reduction varied according to the resistance of the sugarcane varieties. Mishra and Behera (2012) also reported reduction in brix, sucrose and purity of infected canes compared to healthy canes. The extent of reduction was directly correlated with the virulence of the red rot isolate.

Table 3: Score and rating of red rot isolates on various sugarcane varieties

S. No.	Varieties	Red rot score (0-9scale)						Mean
		CF 01	Rating	CF 07	Rating	CF 08	Rating	
1.	BO 153	2.1	MR	2.6	MR	2.8	MR	2.50
2.	BO 91	2.2	MR	2.7	MR	3.2	MR	2.70
3.	BO 154	2.4	MR	3.0	MR	3.4	MR	2.93
4.	CoP 2061	2.8	MR	3.3	MR	3.8	MR	3.30
5.	CoS 8436	6.2	S	6.7	S	7.1	S	6.67
6.	CoSe 95422	6.4	S	6.9	S	7.8	S	7.03
	Mean	3.68		4.20		4.68		

Table 4 – Effect of red rot isolates on cane juice qualities

SL No.	Varieties	Disease score	CF 01			Disease score	CF 07			Disease score	CF 08		
			Brix	Sucrose	Purity (%)		Brix	Sucrose	Purity (%)		Brix	Sucrose	Purity (%)
1.	BO 153	2.10	19.03	16.70	87.77 (69.55)	2.60	18.63	16.00	85.93 (68.11)	2.80	18.10	14.70	81.23 (64.31)
2.	BO 91	2.27	19.27	16.70	86.70 (68.65)	2.77	18.87	15.77	83.58 (66.12)	3.27	18.23	14.23	78.07 (62.05)
3.	BO 154	2.47	19.27	16.37	84.93 (67.24)	3.03	18.77	15.30	81.52 (64.53)	3.40	18.23	13.90	76.21 (60.84)
4.	CoP 2061	2.87	18.87	15.70	83.20 (65.84)	3.37	18.30	14.63	79.97 (63.39)	3.80	17.77	13.30	74.84 (59.89)
5.	CoS 8436	6.23	16.77	12.60	75.17 (60.10)	6.73	15.87	11.47	72.28 (58.22)	7.13	15.37	10.27	66.83 (54.81)
6.	CoSe 95422	6.47	16.57	11.47	69.20 (56.27)	6.93	15.47	10.27	66.36 (54.53)	7.83	15.17	9.17	60.44 (51.01)
	CD (5%)	0.57	0.77	0.79	3.78	0.19	0.77	0.89	3.56	0.81	0.67	0.88	2.64
	CV	8.31	2.33	2.94	3.25	7.56	2.43	3.56	3.17	9.36	2.17	3.89	2.50
	SE(m)±	0.18	0.25	0.25	1.21	0.59	0.25	0.29	1.14	0.25	0.21	0.28	0.85

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